

Chapter 5

Annual Worth Analysis

Annual Worth (AW) analysis is an economic evaluation technique that expresses all cash flows of an alternative as an equivalent uniform annual amount over a specified period. The Annual Worth criterion is mathematically equivalent to the Present Worth (PW) and Future Worth (FW) criteria. Given an interest rate i (typically the MARR) and a planning horizon of n years, the AW is determined directly from PW or FW as:

$$AW = PW \times (A/P, i, n) = FW \times (A/F, i, n) \quad (5.1)$$

Despite this mathematical equivalence, AW analysis is frequently the preferred method in engineering practice because it is easier to compute and offers a direct financial interpretation in terms of annual operational budgets.

5.1 Advantages of AW Analysis

Ease of Understanding The AW value represents the equivalent uniform annual cash flow (expressed in dollars per year) of all estimated receipts and disbursements over the life of an asset. Because corporate budgets, profit-and-loss statements, and financial reports are routinely prepared on an annual basis, AW values are immediately understandable to management and decision-makers without requiring conversion from large present-value lump sums.

Computational Advantage: Single Life Cycle Evaluation Unlike Present Worth analysis—which requires evaluating alternatives over their Lowest Common Multiple (LCM) of lives to satisfy the equal-service requirement—the AW value needs to be calcu-

lated for **only one life cycle** (n years) of each alternative. The resulting AW value for one life cycle is identical to the AW value calculated over 2, 3, or any number of repeated life cycles (including LCM).

Important – Single Life Cycle Evaluation in AW Analysis

With AW analysis, evaluate alternatives over **one life cycle** of each alternative. It is not necessary to extend the cash flow timeline to the LCM of lives.

Proof – Life-Cycle Equivalence ($AW_{1\text{-cycle}} = AW_{\text{LCM}}$)

To understand why evaluating one life cycle is sufficient, consider an alternative with a useful life of n years and an equivalent present worth of PW_1 for its first life cycle. If the alternative is repeated for k life cycles (giving a total planning horizon of $N = k \times n$ years), the total present worth over the k cycles (PW_{LCM}) is:

$$\begin{aligned} PW_{\text{LCM}} &= PW_1 + PW_1(P/F, i, n) + PW_1(P/F, i, 2n) + \cdots + PW_1(P/F, i, (k-1)n) \\ &= PW_1 \left[\sum_{j=0}^{k-1} (1+i)^{-jn} \right] \end{aligned}$$

Using the geometric series summation formula $\sum_{j=0}^{k-1} x^j = \frac{1-x^k}{1-x}$ where $x = (1+i)^{-n}$:

$$PW_{\text{LCM}} = PW_1 \left[\frac{1 - (1+i)^{-kn}}{1 - (1+i)^{-n}} \right]$$

Converting PW_{LCM} to an annual worth over the entire $N = kn$ period gives:

$$\begin{aligned} AW_{\text{LCM}} &= PW_{\text{LCM}} \times (A/P, i, kn) = PW_{\text{LCM}} \left[\frac{i(1+i)^{kn}}{(1+i)^{kn} - 1} \right] \\ &= PW_1 \left[\frac{1 - (1+i)^{-kn}}{1 - (1+i)^{-n}} \right] \left[\frac{i}{1 - (1+i)^{-kn}} \right] \\ &= PW_1 \left[\frac{i}{1 - (1+i)^{-n}} \right] \\ &= PW_1 \times (A/P, i, n) = AW_{1\text{-cycle}} \end{aligned}$$

Thus, $AW_{\text{LCM}} = AW_{1\text{-cycle}}$ identically. This proves that evaluating AW over a single life cycle satisfies the equal-service requirement automatically.

5.2 Fundamental Assumptions of AW Analysis

The validity of comparing alternatives based on a single life cycle rests on three fundamental assumptions:

Definition – Repeatability Assumptions

1. **Equal-Service Requirement:** The service provided by the alternatives will be needed for a period equal to the LCM of their lives or longer (indefinitely).
2. **Identical Replacement:** The selected alternative will be repeated over each future life cycle in exactly the same manner.
3. **Constant Cash Flow Estimates:** The estimated cash flows (initial investment, salvage value, annual revenues, and operating costs) will remain identical in each future life cycle.

If future cash flow estimates are expected to change due to inflation, technological obsolescence, or changing market demand, a fixed study period analysis must be used instead of a single life-cycle AW evaluation.

5.3 Calculation of Capital Recovery (CR) and AW Values

Definition – Annual Worth

The annual worth (AW) value for an alternative comprises two components: capital recovery for the initial investment (P) at a stated interest rate (usually the MARR) and the equivalent annual amount of Operating Cost AOC.

$$AW = CR + AOC$$

Note – Industry Terminology: CapEx and OpEx

In professional engineering and financial modeling, the initial investment is frequently referred to as a **Capital Expenditure (CapEx)**, and the annual operating costs are termed **Operational Expenditures (OpEx)**.

Typically, CR represents an equivalent annual capital cost, while AOC represents Annual Operating Costs (though it can include annual revenues or savings). The total annual amount AOC is determined from uniform recurring costs (and possibly receipts) and nonrecurring amounts.

Definition – Capital Recovery

Capital Recovery (CR) is the **equivalent** annual amount an alternative must earn each year to just recover the initial capital expenditure (CapEx) plus a stated rate of return. Any expected salvage value is considered in the computation of CR^a.

$$CR = -P(A/P, i, n) + S(A/F, i, n)$$

^aIn practice, CR is very sensitive to salvage value estimates. A 20% error in salvage value can shift CR by several percent. Engineers routinely run AW under two or three salvage scenarios before presenting to management (this is a form of sensitivity analysis).

5.4 Decision Guidelines for AW Analysis

The decision guidelines for Annual Worth analysis parallel those of Present Worth analysis:

- **Mutually Exclusive Alternatives:**

- *Revenue Alternatives:* Select the alternative with the **maximum positive** $AW(i = \text{MARR})$. If no alternative yields $AW \geq 0$, select the Do-Nothing (DN) alternative.
- *Cost Alternatives:* Select the alternative with the **minimum annual cost** (numerically least negative or smallest positive cost AW_{cost}). The Do-Nothing option is typically not applicable as the service is mandatory.

- **Independent Projects:**

- Select all independent projects that satisfy $AW(i = \text{MARR}) \geq 0$, provided there are no capital budget constraints.

Example 5.1. Baby Diaper Production Facility Analysis using Annual Worth.

Reconsider the baby diaper production facility decision introduced in Chapter ?? . The firm evaluates three mutually exclusive production line alternatives (Table 5.1) and three independent process improvement projects. The Minimum Attractive Rate of Return (MARR) is 10% per year.

Part 1: Evaluating Mutually Exclusive Alternatives over One Life Cycle Using Annual Worth analysis, we calculate the Capital Recovery (CR) and net AW for **one life cycle** of each alternative ($n_A = 6$, $n_B = 3$, $n_C = 4$).

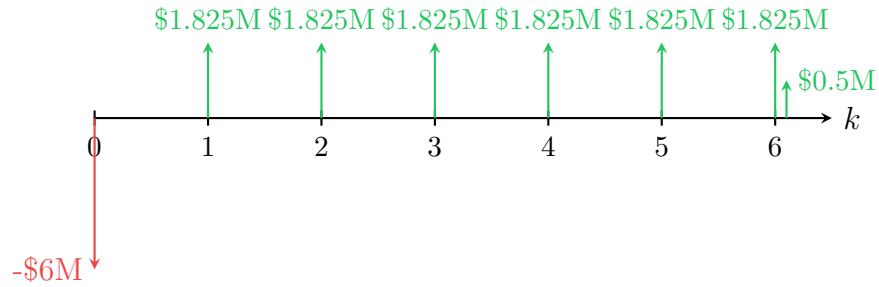
Alternative	Initial Cost (\$)	Annual Revenue (\$)	Salvage Value (\$)	Useful Life n (years)
A (New line)	-6,000,000	1,825,000	500,000	6
B (Used line)	-2,500,000	730,000	200,000	3
C (Co-developed)	-5,000,000	1,095,000	400,000	4

Table 5.1: Summary of mutually exclusive alternatives for the diaper facility.

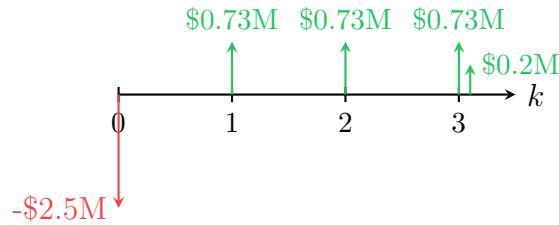
The relevant interest factors at $i = 10\%$ are:

- For $n = 6$: $(A/P, 10\%, 6) = 0.22961$, $(A/F, 10\%, 6) = 0.12961$
- For $n = 3$: $(A/P, 10\%, 3) = 0.40211$, $(A/F, 10\%, 3) = 0.30211$
- For $n = 4$: $(A/P, 10\%, 4) = 0.31547$, $(A/F, 10\%, 4) = 0.21547$

Alt. A



Alt. B



Alternative A ($n = 6$ years):

$$\begin{aligned}
 CR_A &= -6,000,000(A/P, 10\%, 6) + 500,000(A/F, 10\%, 6) \\
 &= -6,000,000(0.22961) + 500,000(0.12961) \\
 &\approx -\$1,312,841
 \end{aligned}$$

$$AW_A = CR_A + 1,825,000 \approx -1,312,841 + 1,825,000 = \$512,159$$

Important – Interpretation of CR and AW

The interpretation of these results is very important.

- **Capital Recovery** ($CR_A \approx -\$1,312,841$): The line must generate at least this much equivalent net revenue every year just to recover the CapEx and the 10% required return.
- **Annual Worth** ($AW_A \approx \$512,159$): After recovering the CapEx and required return, the project generates an **equivalent extra profit** of \$512,159 every single year. Any $AW > 0$ indicates economic viability because it exceeds the minimum required return.

Alternative B ($n = 3$ years):

$$\begin{aligned}
 CR_B &= -2,500,000(A/P, 10\%, 3) + 200,000(A/F, 10\%, 3) \\
 &= -2,500,000(0.40211) + 200,000(0.30211) \\
 &\approx -\$944,864 \\
 AW_B &= CR_B + 730,000 \approx -944,864 + 730,000 = -\$214,864
 \end{aligned}$$

Alternative C ($n = 4$ years):

$$\begin{aligned}
 CR_C &= -5,000,000(A/P, 10\%, 4) + 400,000(A/F, 10\%, 4) \\
 &= -5,000,000(0.31547) + 400,000(0.21547) \\
 &\approx -\$1,491,166 \\
 AW_C &= CR_C + 1,095,000 \approx -1,491,166 + 1,095,000 = -\$396,166
 \end{aligned}$$

Decision: Alternative A yields the highest positive Annual Worth ($AW_A \approx \$512,159$).

Part 2: Comparison with Chapter 5 Present Worth Analysis (LCM = 12 years) In Section ?? of Chapter ??, evaluating these alternatives over the LCM = 12 years required constructing multi-cycle cash flow timelines and discounting reinvestments at years 3, 4, 6, 8, and 9. The resulting present worths over 12 years were:

$$PW_A = \$3,489,696.37, \quad PW_B = -\$1,464,017.41, \quad PW_C = -\$2,699,350.97$$

Converting these 12-year present worth values directly to Annual Worth using $(A/P, 10\%, 12) = 0.14676$:

$$AW_A = 3,489,696.37 \times (A/P, 10\%, 12) = 3,489,696.37 \times 0.14676 \approx \$512,159$$

$$AW_B = -1,464,017.41 \times (A/P, 10\%, 12) = -1,464,017.41 \times 0.14676 \approx -\$214,864$$

$$AW_C = -2,699,350.97 \times (A/P, 10\%, 12) = -2,699,350.97 \times 0.14676 \approx -\$396,166$$

Notice that the AW values derived from the 12-year LCM Present Worth match the single life-cycle AW values **exactly**. This confirms the computational advantage of AW analysis: evaluating one life cycle of 6, 3, and 4 years yields the exact same decision and equivalent annual worth as expanding cash flows over a 12-year LCM horizon.

Part 3: Evaluating Independent Process Improvement Projects The three independent projects evaluated in Chapter ?? have a useful life of $n = 5$ years. The factor $(A/P, 10\%, 5) = 0.26380$.

- **Project 1:** Cost = \$100,000, Net Annual Savings = \$13,000
 $CR_1 = -100,000(0.26380) = -\$26,380$
 $AW_1 \approx -26,380 + 13,000 = -\$13,380 \Rightarrow \text{Reject } (AW < 0)$
- **Project 2:** Cost = \$16,000, Net Annual Savings = \$5,000
 $CR_2 = -16,000(0.26380) \approx -\$4,221$
 $AW_2 \approx -4,221 + 5,000 = \$779 \Rightarrow \text{Accept } (AW \geq 0)$
- **Project 3:** Cost = \$300,000, Net Annual Savings = \$40,000
 $CR_3 = -300,000(0.26380) \approx -\$79,139$
 $AW_3 \approx -79,139 + 40,000 = -\$39,139 \Rightarrow \text{Reject } (AW < 0)$

Decision: Select **Project 2** only, as it is the sole independent project with a positive Annual Worth ($AW_2 \approx \$779$).

5.5 Annual Worth Analysis of Permanent Investments

When evaluating projects that have an indefinite or infinite useful life ($n = \infty$), the evaluation is referred to as **Annual Worth analysis of permanent investments**.

Definition – Perpetuity

A **perpetuity** is an infinite series of equal, periodic cash flows. In engineering economy, analyzing permanent investments (such as infrastructure projects or endowments) often requires treating their operational costs, revenues, or equivalent annualized costs as a perpetuity.

This approach is conceptually identical to the Capitalized Cost (CC) analysis discussed in Section ?? of Chapter ?. Recall that Capitalized Cost is the present worth of a project with an infinite life ($CC = PW_\infty$). The Annual Worth (AW) and Capitalized Cost (CC) are directly related by:

$$AW = CC \times i \quad (5.2)$$

where i is the interest rate (MARR).

To calculate the total net Annual Worth of a permanent investment directly without first finding the Capitalized Cost, each type of cash flow is converted into an equivalent uniform annual amount:

- **Initial Cost (P at $t = 0$):** The equivalent annual cost of recovering an initial investment P in perpetuity is:

$$A_P = P(A/P, i, \infty) = P \cdot i \quad (5.3)$$

- **Recurring Cash Flow (R every n_R years):** For a periodic recurring cash flow R that repeats every n_R years (with the first occurrence at Year n_R), its equivalent uniform annual worth A_R over each n_R -year life cycle is:

$$A_R = R(A/F, i, n_R) = R \left[\frac{i}{(1+i)^{n_R} - 1} \right] \quad (5.4)$$

- **Non-Recurrent Cash Flow (C at Year n_C):** For a one-time non-recurrent cash flow C occurring at Year n_C , it is first discounted to the present and then multiplied by i to convert it into a perpetual annual amount:

$$A_C = C(P/F, i, n_C)(A/P, i, \infty) = \frac{C \cdot i}{(1+i)^{n_C}} \quad (5.5)$$

- **Uniform Annual Cash Flow (A):** Any recurring uniform annual cash flow (such

as annual maintenance or operating costs) is already an annual amount and requires no transformation.

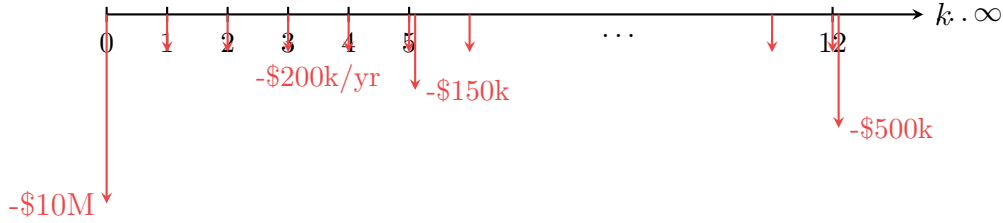
Example 5.2. Annual Worth of a Permanent Irrigation Dam.

A government agency is planning a permanent hydroelectric and irrigation dam ($n = \infty$) with a MARR of 6% per year. The estimated costs are:

- Initial construction cost at $t = 0$: \$10,000,000
- Routine annual operation and maintenance (AOC): \$200,000 per year
- Non-recurrent structural inspection cost at Year 5: \$150,000
- Recurring major gate overhaul every 12 years: \$500,000 (first overhaul at Year 12)

Determine the total equivalent annual cost (AW_{cost}) of the dam in perpetuity.

Solution:



We convert each cost component into an equivalent annual amount using Equations 5.3–5.5 at $i = 6\%$:

1. Initial Construction Cost ($P = \$10,000,000$):

$$A_P = -10,000,000 \times 0.06 = -\$600,000 \text{ per year}$$

2. Routine Annual Maintenance ($AOC = \$200,000$):

$$A_{\text{maint}} = -\$200,000 \text{ per year}$$

3. Non-Recurrent Inspection ($C = \$150,000$ at $n_C = 5$):

$$A_C = \frac{-150,000 \times 0.06}{(1 + 0.06)^5} = \frac{-9,000}{1.33823} \approx -\$6,725 \text{ per year}$$

4. **Recurring Major Overhaul** ($R = \$500,000$ every $n_R = 12$ years):

$$\begin{aligned} A_R &= -500,000(A/F, 6\%, 12) = -500,000 \left[\frac{0.06}{(1 + 0.06)^{12} - 1} \right] \\ &= -500,000(0.05928) \approx -\$29,640 \text{ per year} \end{aligned}$$

Summing all annual equivalent costs gives the total perpetual annual cost:

$$\begin{aligned} AW_{\text{cost}} &= A_P + A_{\text{maint}} + A_C + A_R \\ &\approx -600,000 - 200,000 - 6,725 - 29,640 \\ &= -\$836,365 \text{ per year} \end{aligned}$$

Thus, the permanent dam requires an equivalent annual expenditure of approximately \$836,365 per year in perpetuity.

In simpler words, if the government agency budget allows spending \$836,363.84 every year, they will be able to build the dam, maintain it, inspect it, and overhaul it forever without ever running out of money.

Note – Clarity of Annual Worth

The use of annual worth of the project highlights the ease of understanding of the cost associated of a project.